A Prototypical Metric Learning Approach for Open-Set Semantic Segmentation on Remote Sensing Images

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***Abstract*— Semantic segmentation has received wide attention as a feasible solution to effectively interpret the information in remote sensing images. Solutions are typically built with a static closed-set perception, where all labels are known a priori. However, in real-world applications, such as remote sensing images, one has to handle objects from unknown classes. Open-set semantic segmentation (OSSS) is an approach that incorporates open-set perception into semantic segmentation, allowing the recognition of unknown classes of objects. Different studies have explored the use of OSSS in remote sensing images. However, their performance is limited due to the poor and overlapped representation of the features extracted from images. This results in an embedding space with low discrimination among the classes. This article introduces a novel loss function called prototypical triplet loss, which uses prototype representation and metric learning to improve open-set recognition. In addition, two open-set classifiers, one based on principal components and the other on prototypical distance, were also proposed once they took advantage of the features obtained by the prototypical triplet loss. Experiments were done with two public remote sensing image datasets: Vaihingen and Potsdam. The results demonstrate that the proposed methods improve OSSS compared to other state-of-the-art approaches. These results reinforce the importance of this type of approach, enabling applications in real systems that require open-set recognition. All codes are freely available at https://github.com/Brilhador/tgrs2023 to foster further research in this area.**

***Index Terms*— Metric learning, open-set, prototype representation, remote sensing, semantic segmentation.**

I. Introduction

THE development of new aerial technologies for image acquisition has facilitated the collection of many high-quality remote sensing images, making it possible to observe and study various phenomena on the Earth's surface, both on a small and large scale. In this sense, there is a crucial need to develop tools for effectively interpreting the information in remote sensing images [1]. Furthermore, there is a growing interest in building interfaces that enable the mapping and monitoring of cities, regions, countries, or even entire continents [2]. Developing solutions to achieve these goals is currently a significant focus. In this sense, semantic segmentation has received wide attention as a feasible solution, being able to label each pixel present in the image with a known label [3], [4].

Semantic segmentation solutions are typically built on a static closed-set perception, where all labels are known a priori [5], [6], [7]. This closed-set approach assumes that the training and testing pixels belong to the same label space known known classes (KKCs), meaning that the train and test sets contain the same classes. However, this assumption does not hold in real-world scenarios, especially in earth observation applications. During the prediction phase, the model may face pixels from classes not seen during the training phase unknown unknown classes (UUCs). This may happen in large images, where many classes are missed to be labeled during training [2]. In such cases, closed-set semantic segmentation methods are susceptible to errors when dealing with UUCs. During inference, the model may wrongly recognize samples from UUCs as one of the KKCs, leading to misclassifications and limiting the usability of these approaches in real-world earth observation applications.

An effective way to overcome these limitations is to build image recognition systems based on dynamic and open-set perception. These systems are designed to handle objects from UUCs commonly encountered in real-world applications [8]. Open-set semantic segmentation (OSSS) is an approach that incorporates open-set perception into semantic segmentation. Unlike closed-set semantic segmentation, OSSS must correctly classify samples belonging to KKCs while rejecting those belonging to UUCs.

Different studies have explored the use of OSSS in remote sensing images [2], [9], [10], [11], [12], [13]. However, these studies limit themselves to using the successful and widely employed cross-entropy loss. This loss function measures the disparity between the predicted values and the ground truth (GT), guiding the model's learning process based on labeled data. While the outcomes of these studies are promising, their performances are limited due to the low representation of the features obtained. This results in an "irregular" embedding space with low discrimination among the classes—the term "irregular" suggests undefined or overlapping decision boundaries between classes. Recent works [14], [15] show that metric learning can improve open-set recognition by obtaining more discriminative features and creating an "ideal"

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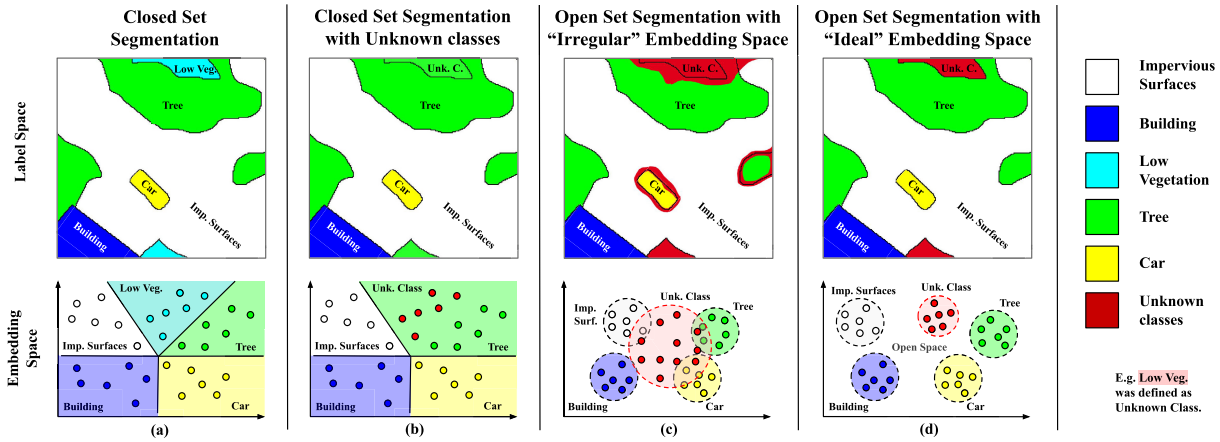


Fig. 1. Graphical representation of closed-set and open-set problem settings in dense labeling scenarios. The label space represents the predictions at the pixel level, and the embedding space refers to a subset of pixel samples in a 2-D manifold separated by labels and decision boundaries for each class. (a) Closed-set without the presence of UUCs. (b) Closed-set with the presence of UUCs. These UUCs result in misclassification, being segmented as KKCs. (c) Open-set segmentation with "irregular" embedding space. The term "irregular" suggests undefined or overlapping decision boundaries between classes. (d) Open-set segmentation with "ideal" embedding space. The term "ideal" refers to well-defined decision regions between the classes and with enough space (open space) between them to include new classes [8], [15].

embedding space. Such an "ideal" space should tightly cluster KKCs while keeping UUCs far from these clusters [8], [15]. Fig. 1 illustrates the idea of "irregular" and "ideal" spaces.

Hence, this article addresses the limitations of open-set recognition in semantic segmentation due to low discrimination features. To address this issue, we propose a new loss function called prototypical triplet loss, which uses a prototype representation and metric learning to improve open-set recognition. Semantic segmentation models trained with this loss function tend to learn more discriminative features and, consequently, construct well-controlled embedding spaces closer to the ideal embedding space. Furthermore, this article proposes two open-set classifiers, one based on principal components and the other on prototypical distance. These classifiers benefit from the well-controlled embedding space produced by the proposed prototypical triplet loss to perform OSSS.

The results of this article demonstrated that the proposed approach improves OSSS in remote sensing images. The main contributions of our work can be summarized as follows.

1) A novel loss function called prototypical triplet loss was developed using prototype representation and metric learning. This loss incorporates global (batch-level) and local (image-level) attention mechanics to improve the optimization of the embedding space while dealing with imbalanced data and uncertainty boundary regions between objects.
2) An online triplet mining approach based on confusion matrix rules was introduced for global and local attention mechanics, providing a simple and efficient method for triplet selection in the context of semantic segmentation.
3) Two open-set classifier approaches: i) generalized open principal components scoring with incremental learning (OpenIPCS) called GOpenIPCS and ii) intuitive approach based on prototypical distance strongly related to the discriminative features generated by the proposed prototypical triplet loss called POpenDist.
4) Detailed and in-depth comparisons of state-of-the-art loss functions and open-set classifiers demonstrate the effectiveness of the proposed approach for OSSS on remote sensing images.

This article is organized as follows. Section II presents related works, mainly including OSSS and prototypical metric learning. Section III describes in detail the proposed methods. Section IV defines the configurations and details of the experiments. Section V presents the obtained results and their discussion. Finally, Section VI presents the conclusion and future research directions.

II. Related Work

In this section, we review the related works to OSSS on remote sensing images and prototypical metric learning.

A. Open-Set Semantic Segmentation

In recent years, deep learning has been successfully applied to remote sensing images [16], [17]. Nevertheless, these methods are unsuitable for open-set scenarios, which are common in real-world computer vision applications. This is because the closed-set perception fails when unknown classes from training are found in the test phase [12]. Consequently, several approaches have been developed to address the challenges associated with open-set classification for remote sensing [18], [19]. However, most of these methods focus on the task of image classification. This article focuses on semantic segmentation or pixel-wise classification, which presents other specific challenges, such as small objects and uncertainty boundary regions [10], [12].

In OSSS in remote sensing images, [9], [11] adapted the OpenMax layer introduced for the image classification problem [20] in an OSSS method. This approach replaces the last activation layer of a semantic segmentation network, usually the SoftMax layer, with the OpenMax layer, estimating the input probability and labeling it as KKCs or UUCs. However, this method has a significant drawback of being

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extremely inefficient during both training and testing [13]. Open fully convolutional network (OpenFCN) was proposed by Oliveira et al. [2] as an alternative to approaches relying on the OpenMax layer. In OpenFCN, the closed-set network is trained in a fully convolutional (FC) manner. However, it presents little performance improvements. On the other hand, the open principal component scoring with incremental training (OpenIPCS) [2] effectively addressed the limitations of OpenMax-based approaches. This method utilizes intermediate feature activations from closed-set networks to model Gaussian distributions in a low-dimensional projection, deriving a likelihood score for unknown pixels. OpenIPCS demonstrates enhanced efficiency during training and testing, resulting in significant performance gains. At last, Nunes et al. [13] proposes an end-to-end FC method called conditional reconstruction for open-set segmentation (CoReSeg), which uses an autoencoder architecture to generate open-set predictions based on pixel location error. However, it is well known that autoencoder-based approaches suffer from several problems, especially attention to overfitting and insufficient accuracy [21]. In contrast to previous approaches, we focus on improving class discriminability using a prototypical metric learning network to build a well-controlled embedding space.

Metric learning has received significant attention for addressing the open-set problem and is being studied in various fields such as image classification [8], semantic segmentation [7], [22], and zero-shot segmentation [23]. This work introduces a novel prototypical metric learning approach that incorporates global and local attention strategies to enhance the optimization of the embedding space while handling imbalanced data and uncertain boundary regions between objects. Additionally, we introduce a generalized open-set classifier approach based on OpenIPCS called GOpenIPCS that benefits from learned embedding space. As shown by Shwartz-Ziv and Tishby [24], the deeper the layers of a convolutional neural network (CNN), the closer the embedding vector will be to the label space. Therefore, the performance of OpenIPCS may be reduced due to the noise or unnecessary information in the intermediate layers. Thus, GOpenIPCS only concentrates on the last two layers of the CNN, which are efficiently coupled to different architectures to generate an open-set prediction. As detailed in Section V, our approach demonstrates significant advancements in OSSS on remote sensing images.

B. Prototypical Metric Learning

Prototypical learning [25], [26] aims to associate each class with a representation or prototype. Observations are then classified based on the proximity to the nearest prototype. In contrast, metric learning is an approach focused on learning a distance metric that reinforces similarity between objects in the embedding space, ensuring that instances of the same class are close while instances of different classes are positioned far apart.

Although such paradigms have been successfully employed separately for distinct tasks, including image classification [16], [27], semantic segmentation [28], [29], and few-shot learning [3], [17], there is a notable scarcity of works that integrate prototypical and metric learning in the context of OSSS. Specifically, only the work by Cen et al. [7] has explored the fusion of prototypical learning and metric learning, focusing on enhancing the autonomous driving context. This work suggests employing prototypes for all classes and utilizing metric learning to improve the embedding space. The aim is to enhance the compactness within the same class and separability between different classes. However, it is essential to note that this methodology does not impose a minimum distance constraint between distinct classes, overlooks considerations of class imbalance, and does not incorporate global and local attention mechanisms to increase class discrimination.

Unlike this previous work, we proposed a new loss function based on prototypical metric learning that defines a minimum margin between KKC to enhance the models' discriminative capability. Moreover, our approach deals with imbalanced data and incorporates global (batch-level) and local (image-level) attention mechanics to improve the optimization of the embedding space. The results, which are elaborated in Section V, demonstrate the effectiveness of our proposed approach.

III. Proposed Methods

This section outlines the proposed methods as follows. Section III-A provides a detailed description of the prototypical triplet loss, briefly illustrated in Fig. 2. Section III-B presents the two proposed open-set classifiers using principal components or prototypical distance.

A. Prototypical Metric Learning Network

CNNs for semantic segmentation [6] can be divided into two parts: a feature extractor, which includes several convolution layers followed by max-pooling and activation function; and a linear classifier $\boldsymbol{f} = \boldsymbol{W}^{\top}\boldsymbol{x}+\boldsymbol{b}$ in the last fully connected layer. Such layer processes the feature vector $\boldsymbol{x}$ of the penultimate layer for obtaining the embedding vector $\boldsymbol{f} \in \mathbb{R}^{C}$ of each pixel of the input image, where C represents the number of classes. Actually, this learnable classifier is unsuitable for open-set recognition because it assigns all feature space to KKCs and leaves no space for UUCs [7]. To attenuate this limitation, Cen et al. [7], Yang et al. [31] proposed a metric learning strategy, replacing this classifier by the Euclidean distance representation of the embedding vectors for each class prototypical anchor $m_y|y \in \{1, 2, \ldots, C\}$ belonging to KKCs. In contrast to those approaches, here we propose a novel loss function called prototypical triplet loss, defined as

$$\mathcal{L}_{\text{PTL}} = \lambda_g \mathcal{L}_{\text{PTL}}^{\text{Global}} + \lambda_l \mathcal{L}_{\text{PTL}}^{\text{Local}} \tag{1}$$

in which λ_g and λ_l represent the weighting hyperparameters of the loss functions.

The main goal of the $\mathcal{L}_{\text{PTL}}$ is to construct a well-controlled embedding space to enhance OSSS [7]. While handles other challenges inherent to semantic segmentation, such as class imbalance (small objects) and uncertainty boundary regions, which can negatively impact OSSS [10], [12]. To achieve this, loss is divided into two: one focused on global attention and the other on local attention, which will be detailed below.

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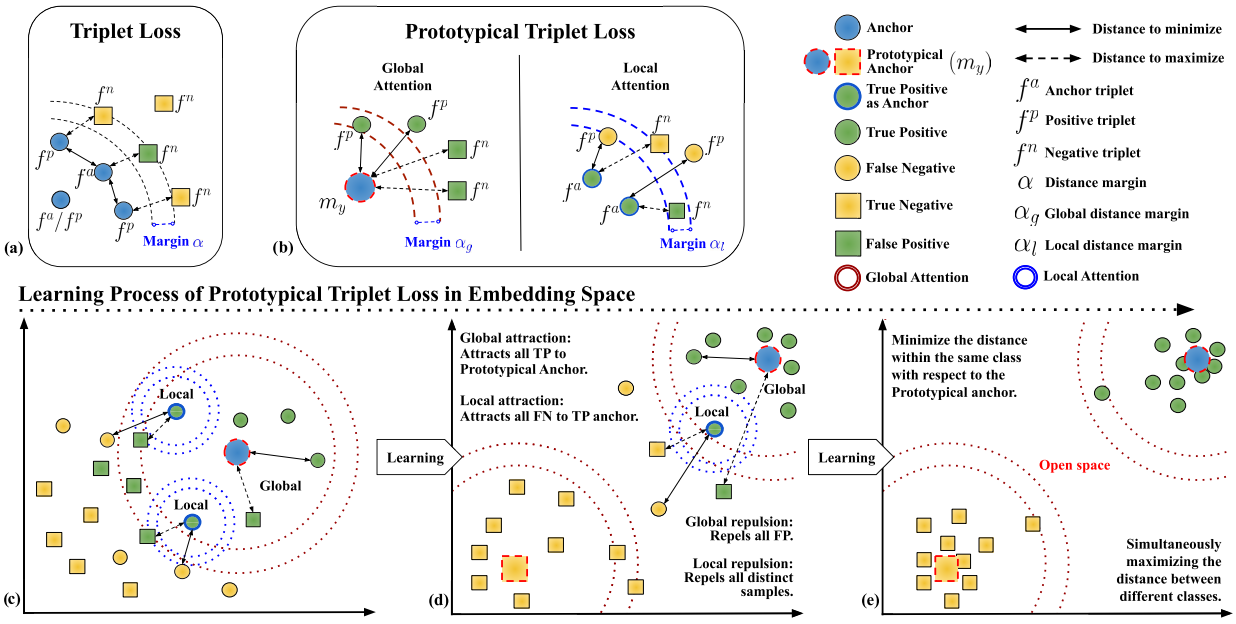


Fig. 2. (Top) Graphical representation of online triplets selection strategies. (Bottom) Illustration of the learning process of $\mathcal{L}_{\text{PTL}}$ proposed in this article. (a) Online triplet selection for image classification [30]. (b) Online triplet selection used by $\mathcal{L}_{\text{PTL}}$ proposed by this article. The global attention mechanism works by pulling TP samples to the prototypical anchor, while pushing FP samples further away by a minimal margin. Notice that the prototypical anchors are static and defined prior to the learning process. The local attention uses the TP samples as anchors, attracting closer the FN samples, while pushing the FP samples by a minimal margin. (c) During the initial learning stage, the samples in the embedding space are randomly distributed. As a result, the global attention mechanism exerts a stronger influence since the samples are more distant from the prototypical anchor, accelerating the convergence of the model in the initial steps. (d) As the training progresses, the effect of the global attention mechanism decreases, while the importance of the local attention mechanism increases. This transition is driven by the increasing difficulty in discerning the differences among closely related samples that belong to different classes. In addition, $\mathcal{L}_{\text{PTL}}$ provides attention forces. Specifically, TP samples gradually move toward the prototypical anchors, while FN samples are attracted to the TP samples. In contrast, the TN and FP samples are pushed away simultaneously. (e) Finally, the expected result is to minimize the distance within the same class concerning the prototypical anchor while maximizing the distance between different classes, preserving an open space between the classes to include the new classes in the future.

1) Global Attention Loss Function: The global attention loss function ($\mathcal{L}_{\text{PTL}}^{\text{Global}}$) is a modification of the triplet loss [30], in which the objective is to push away the pixel-level negative triplets f^n from the anchor f^a by a distance margin $\alpha_g > 0$ compared to the pixel-level positive triplets f^p

$$\|f^a - f^p\|_2^2 + \alpha_g \leq \|f^a - f^n\|_2^2. \tag{2}$$

For instance, as shown in Fig. 2(b), f^p is expected to stay closer to the anchor f^a to f^n. Thus, to enforce this constraint, the triplet loss is defined as a relaxation of (2), defined as the minimization of the following equation:

$$\mathcal{L}_{\text{triplet}} = \left[||f^a - f^p||_2^2 + ||f^a - f^n||_2^2 + \alpha\right]_+ \tag{3}$$

where operator $[.]_+ = \max(0, .)$ ensuring the loss is non-negative.

The proposed $\mathcal{L}_{\text{PTL}}^{\text{Global}}$ makes two main modifications to the triplet loss presented in (3). First, it replaces the triplet loss anchors f^a with predefined prototypical anchors m_y. These m_y are evenly distributed in the embedding space and are defined as static vectors of the same size as the last layer of the CNN. For each semantic class y, the m_y has the yth element attributed with a magnitude value of T. This approach aims to ensure a well-controlled embedding space with spatially and uniformly distributed m_y as proposed in [7] and [15]. Second, (3) is computed for each class independently and then averaged across these triplets. This method aims to equalize the importance of different classes during training, thereby addressing the issue of class imbalance [32]. Thus, the $\mathcal{L}_{\text{PTL}}^{\text{Global}}$ is defined as follows:

$$\mathcal{L}_{\text{PTL}}^{\text{Global}} = \frac{1}{\hat{C}} \sum_{k=0}^{\hat{C}} \sum_{j=0}^{\hat{C}_t} \left[\left\| m_{y_k} - f_j^p \right\|_2^2 - \left\| m_{y_k} - f_j^n \right\|_2^2 + \alpha_g \right]_+ \tag{4}$$

where $\hat{C}_t$ is the number of available triplets in batch for each class, $k \in \{1, 2, \ldots, \hat{C}\}$, $\hat{C}$ is the number of available classes in the batch, m_{y_k} refers to prototypical anchor, f^p is the pixel-level positive triplets input of the same class as m_{y_k}, f^n is the pixel-level negative triplets input of a different class from m_{y_k}, and α_g is a global margin between f^p and f^n. The number of f^p and f^n must have the same amount of pixel embeddings. To achieve this balance, a limited number of samples were randomly selected, constrained by the minimum count between f^p and f^n. Note that, by replacing f^a by m_y in (3), the $\mathcal{L}_{\text{PTL}}^{\text{Global}}$ can be represented as follows:

$$\mathcal{L}_{\text{PTL}}^{\text{Global}} = \frac{1}{\hat{C}} \sum_{k=0}^{\hat{C}} \sum_{j=0}^{\hat{C}_t} \mathcal{L}_{\text{triplet}}. \tag{5}$$

In addition to these modifications, we propose an efficient online triplet mining approach based on a confusion matrix. The cubical nature of the triplet loss [33] makes it unsuitable for semantic segmentation, and selecting effective triplets is challenging because random choices do not ensure model

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convergence [34]. Our work introduces a simple and efficient online triplet mining method to address this limitation.

The proposed online triplet mining strategy to $\mathcal{L}_{\text{PTL}}^{\text{Global}}$ was adopted based on the confusion matrix of the model predictions into the batch, selecting the true positive (TP) as positive triplets and false positives (FP) as negative triplets. The intuition behind this selection is that the FPs are the hard-negative samples [35], which can accelerate the learning process while obtaining more discriminative models [36]. On the other hand, the false negatives (FN) were not selected as positive triples because they increased the difficulty of the learning process, preventing the model from converging. An overview of the proposed online triplet selection strategy can be seen in Fig. 2.

2) Local Attention Loss Function: While $\mathcal{L}_{\text{PTL}}^{\text{Global}}$ focuses on handling pixel embeddings at the batch-level and ensuring a well-controlled embedding space, the local attention loss function ($\mathcal{L}_{\text{PTL}}^{\text{Local}}$) addresses image-level challenges, particularly with uncertain boundary regions, as discussed in Section V-C. Unlike $\mathcal{L}_{\text{PTL}}^{\text{Global}}$, prototypical anchors are not used, and TPs are defined as anchors to represent the local characteristics of each class within the image. FNs are used as positive triplets, while true negatives (TN) and FPs are used as negative triplets. This approach effectively addresses uncertainties and refines local regions of the image. Similar to the $\mathcal{L}_{\text{PTL}}^{\text{Global}}$, the $\mathcal{L}_{\text{PTL}}^{\text{Local}}$ calculates the triplet loss for each image and averages these losses across the batch, as defined in the following equation:

$$\begin{aligned}\mathcal{L}_{\text{PTL}}^{\text{Local}} &= \frac{1}{\check{C}}\sum_{k=0}^{\check{C}}\sum_{j=0}^{\check{C}_t}\left[\left\|f_j^a - f_j^p\right\|_2^2 - \left\|f_j^a - f_j^n\right\|_2^2 + \alpha_l\right]_+ \\ &= \frac{1}{\check{C}}\sum_{k=0}^{\check{C}}\sum_{j=0}^{\check{C}_t}\mathcal{L}_{\text{triplet}} \end{aligned} \tag{6}$$

where $\check{C}_t$ is the number of triplets available in the image for each class, $k \in \{1, 2, \ldots, \check{C}\}$, and $\check{C}$ is number of available classes in the image. Algorithm 1 shows the process to compute the $\mathcal{L}_{\text{PTL}}$.

Algorithm 1 $\mathcal{L}_{\text{PTL}}$ on one batch of images

Input: Batch of images (X)
Input: Ground truth (Y)
Input: Prototypical anchors (m_y)
Input: The weighting hyperparameters λ_g and λ_l
Output: $\mathcal{L}_{\text{PTL}}$

Compute the pixel embeddings $\boldsymbol{f}$ by forward propagation;
Compute the predictions $\hat{Y}$ by argmax($\boldsymbol{f}$);
$\hat{C}_t = 0$;
$\text{sum}_{global} = 0$;
Foreach: $k \in \hat{C}$ **do**
  Select triplets f^b, and f^n of k_X comparing $\hat{Y}$ and Y;
  if $f^b == 0$ or $f^n == 0$ **then**
    continue;
  end if
  sum_{global} += Compute the Eq. 3 (replace f_a by m_y);
  $\hat{C}_t = \hat{C}_t + 1$;
end foreach
$\mathcal{L}_{triplet}^{global} = \text{sum}_{global} / \hat{C}_t$;
$\check{C}_t = 0$;
$\text{sum}_{local} = 0$;
Foreach: $i \in X$ **do**
  Foreach: $k \in \hat{C}$ **do**
    Select triplets f^a, f^b, and f^n of k_i comparing $\hat{Y}$ and Y;
    if $f^b == 0$ or $f^n == 0$ **then**
      continue;
    end if
    sum_{local} += Compute the Eq. 3;
    $\check{C}_t = \check{C}_t + 1$;
  end foreach
end foreach
$\mathcal{L}_{triplet}^{\text{Local}} = \text{sum}_{local} / \check{C}_t$;
$\mathcal{L}_{\text{PTL}} = \lambda_g \mathcal{L}_{\text{PTL}}^{\text{Global}} + \lambda_l \mathcal{L}_{\text{PTL}}^{\text{Local}}$ (Eq. 1)
return $\mathcal{L}_{\text{PTL}}$

B. Open-Set Classifier for Semantic Segmentation

In semantic segmentation networks, the embedding vectors $\boldsymbol{f} \in \mathbb{R}^C$ are commonly normalized using the softmax function into a probability distribution for each class y, so as to perform the final classification of each pixel. Thus, to obtain the probability of $\boldsymbol{f}$ to belong to class y, one can apply the softmax function defined as follows:

$$p_y(\boldsymbol{f}) = -\log\frac{\exp(f_y)}{\sum_{c=1}^{C}\exp(f_c)}. \tag{7}$$

Therefore, the final classification of each pixel is defined as $\hat{Y}_f^{\text{close}} = \text{argmax } p_y(\boldsymbol{f})$. However, this learnable classifier cannot recognize UUC, making it unsuitable for open-set recognition, since it assigns all features to KKCs [7]. Thus, open-set classifiers are necessary to classify KKCs, while recognizing UUCs accurately. Previous studies have investigated this development in the semantic segmentation task [2], [7], [13]. Based on these works, one can define the general open-set classifier as follows:

$$\hat{Y}_f^{\text{open}} = \begin{cases} C_{\text{UUC}} & \max(p_y(\boldsymbol{f})) \leqslant \lambda_{\text{out}} \\ \hat{Y}_f^{\text{close}} & \max(p_y(\boldsymbol{f})) > \lambda_{\text{out}} \end{cases} \tag{8}$$

where C_{UUC} denotes the UUC and λ_{out} is the threshold to determine the UUC pixels. Therefore, we propose two open-set classifier methods that benefit from the embedding vectors provided by the $\mathcal{L}_{\text{PTL}}$, detailed as follows.

1) Open-Set Classifier Using Prototypical Distance: Since long ago, distance-based classifiers, such as the K-nearest neighbor (KNN) classifier, have been widely used in pattern recognition [37]. These techniques define the classification of a given data sample based on its proximity to other labeled points in a multidimensional space. Based on this premise, we propose an open-set classifier based on the distances between samples and the prototypical anchors called open prototypical distance (POpenDist). For distinguishing between KKCs and UUCs, we use the minimum distance between samples and the prototypical anchors $\min(d(f))$. If the samples

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have a $\min(d(f))$ above a predefined threshold (λ_{dist}), they are labeled as UUC. Otherwise, the labels are assigned in the closed-set segmentation, according to the following:

$$\hat{Y}_f^{\text{popen}} = \begin{cases} C_{\text{UUC}} & \min(d(f)) > \lambda_{\text{dist}} \\ \hat{Y}_f^{\text{close}} & \min(d(f)) \leqslant \lambda_{\text{dist}}. \end{cases} \tag{9}$$

This open-set classification approach takes advantage of the label space induced by the $\mathcal{L}_{\text{PTL}}$, in which the KKC samples are projected closer to their corresponding prototypical anchors. At the same time, the UUC samples are projected toward the open space and, therefore, further away from the prototypical anchors.

2) Open-Set Classifier Using Principal Components: A recent study [2] introduced an open-set classifier method called open principal component scoring with incremental learning (OpenIPCS). This method is designed for OSSS and is based on principal components analysis (PCA) applied to the internal feature vectors of CNNs. The strategy behind this open-set classifier involves training a generative model G using the validation images. G is used as a probability density estimator with Gaussian priors. In open-set recognition, a threshold (λ_g) is applied to the likelihoods assigned to the test pixels belonging to KKCs. These likelihoods are obtained from the difference between the feature vector and the projected feature vector. The projection step involves extracting principal components and reconstructing the feature vector using G. This procedure leads pixels belonging to UUCs to have high difference values since they do not belong to the training dataset. Conversely, pixels KKCs have low difference values. Therefore, open-set recognition from OpenIPCS is achieved as follows:

$$\hat{Y}_f^{\text{gopen}} = \begin{cases} C_{\text{UUC}} & \text{abs}\left(\hat{f}_{\text{test}} - \hat{f}_{\text{test}}^G\right) > \lambda_g \\ \hat{Y}_f^{\text{close}} & \text{abs}\left(\hat{f}_{\text{test}} - \hat{f}_{\text{test}}^G\right) \leqslant \lambda_g. \end{cases} \tag{10}$$

OpenIPCS only uses multiple-level internal network layers based on an FC network (FCN) decoder. This design constraint limits the applicability of the method to different network architectures. In response to these limitations, this article introduces a more generalized version of OpenIPCS, named GOpenIPCS. GOpenIPCS takes a different approach by focusing on the last and penultimate layers of CNNs. This choice is based on the findings from [24], who suggested that the deeper layers are closer to the label space. Thus, these layers contain more semantically significant information. In contrast, the shallower layers may contain noisy information that can interfere with classification. Therefore, GOpenIPCS omits other network layers, making it a more versatile solution easily integrated into various network architectures. OpenIPCS avoided using the last layers of the network due to uncertainty boundaries present in these layers [2]. However, the proposed $\mathcal{L}_{\text{PTL}}$ strategy is a more effective way to address these uncertainties while improving the performance of the open-set classifier.

IV. Experimental Setup

This section outlines the experimental setup for evaluating the proposed methods and other baselines in closed- and open-set scenarios. Section IV-A details the datasets. Next, Sections IV-B and IV-C describe the data sampling and preprocessing, respectively. Section IV-D defines the settings to obtain the best model in the closed-set. Then, Section IV-E briefly describes the training flow of the OSSS methods. Section IV-F describes the step of defining the cutoff thresholds to acquire open-set predictions. Finally, Section IV-G defines the evaluation metric. Fig. 3 provides an overview of the proposed method. Each stage of Fig. 3 is detailed as follows.

A. Datasets

To evaluate the effectiveness of the proposed methods, experiments were performed on two 2-D urban scene semantic labeling datasets: Vaihingen and Potsdam. These datasets have pixel-level labeling and are available from the International Society of Photogrammetry and Remote Sensing (ISPRS).

1) Vaihingen: This dataset consists of 33 true orthophoto (TOP) images, out of which 16 images have the GT publicly available. Each image has a spatial resolution of 9 cm per pixel and several resolutions. The images have four channels: near-infrared, red, green (IRRG), and normalized digital surface model (NDSM). Based on [4], we excluded the NDSM channel from the experiments because many remote sensing datasets do not have DSM and NDSM. These images are labeled into six categories for semantic segmentation: impervious surfaces, buildings, low vegetation, trees, cars, and clutter. The clutter class was ignored in our experiments, as it contains a relatively small number of samples. Following [2], we selected five images for testing: 11, 15, 28, 30, and 34, and the remaining 11 images were used for training. The images were cropped into mini-patches of 224 $\times$ 224 pixels, as can be seen in Fig. 4. Any missing pixels resulting from the cropping process were filled in with black pixels, which are disregarded in our experiments.

2) Potsdam: This dataset contains 38 images with the IRRG and NDSM channels and public GT. As mentioned before, the NDSM channel was not used. Each image has a spatial resolution of 5 cm and 6000 $\times$ 6000 pixels. This dataset has the same categories as the Vaihingen dataset, and the clutter class was not considered in our experiments. Following [2], we selected seven images for testing: 2_11, 2_12, 4_10, 5_11, 6_7, 7_8, and 7_10, and the remaining 31 images were used for training. Similar cropping and missing pixels treatments were used here as the Vaihingen dataset.

B. Data Sampling

According to [2], one can define nine experimental conditions where KKCs are marked as UUCs and are ignored during training. These nine combinations of UUCs are as follows: 1) impervious surfaces; 2) buildings; 3) low vegetation; 4) high vegetation; 5) car; 6) low vegetation and high vegetation; 7) buildings and car; 8) impervious surfaces and car; and 9) impervious surfaces and buildings.

Next, the training data for each experiment was divided into training and validation sets using the k-fold cross-validation method, with a k value of 3 for each experimental condition.

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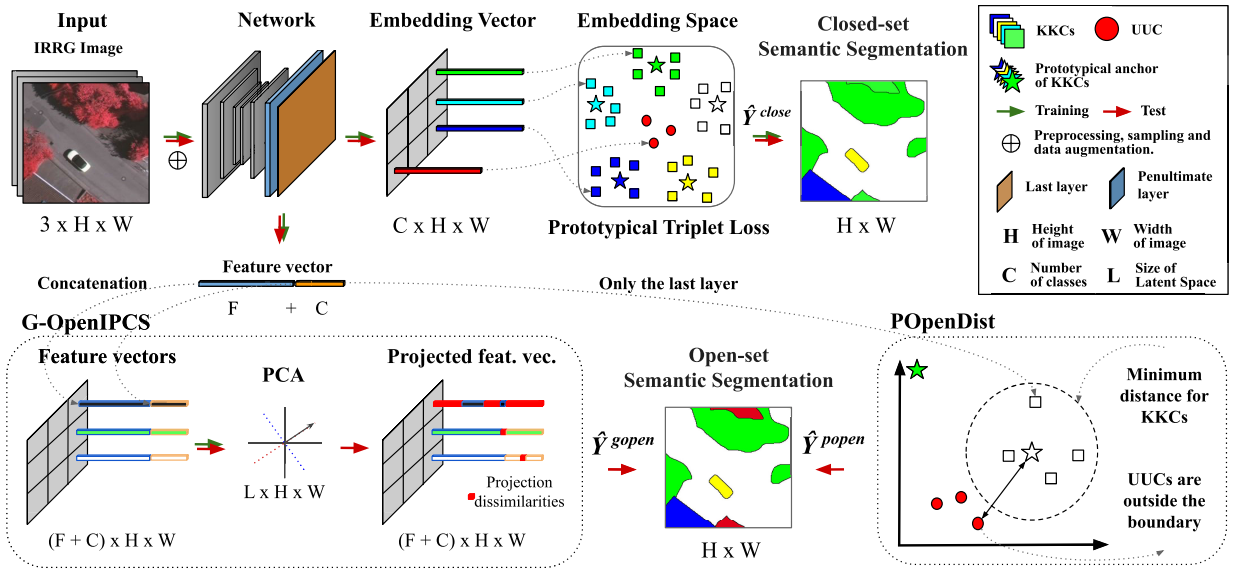


Fig. 3. Overview of the proposed methods. Closed-set segmentations are obtained from models trained by different loss functions. The $\mathcal{L}_{\text{PTL}}$ expands the decision boundaries between classes to improve class discrimination while dealing with the problem of imbalance between classes and uncertainty boundary regions. The open-set segmentation process is highlighted within the dotted block. The proposed GOpenIPCS recognizes UUC pixels, learning on discrepancies between feature vectors. This approach utilizes the last two layers of the segmentation model. The proposed POpenDist recognizes the UUC pixels on distances from prototypical anchors and only uses the last layer.

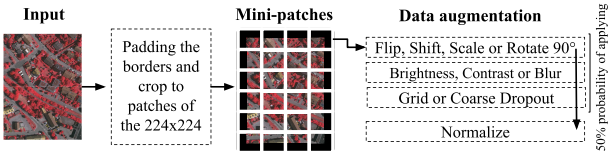


Fig. 4. Padding, crop, and data augmentation.

The choice of a small value for k is due to the long training time required for semantic segmentation models. According to this selection method, 2/3 of the images from the training data are randomly selected into the training set, while the remaining 1/3 are assigned to the validation set.

C. Data Preprocessing

The preprocessing pipeline begins with padding and cropping the images to 224 × 224 pixels, as proposed in [38]. Next, data augmentation transformations are divided into three blocks, with a 50% probability of applying one transformation within each block. The first block includes horizontal and vertical flipping, shifting (limited to 0.2), scaling (limited to 0.2), or rotating the image by 90°. The second block applies one of the following transformations: random brightness adjustment, contrast adjustment (both limited to 0.3), or Gaussian blur (with a kernel size up to 5). The third block involves either grid dropout or coarse dropout, limited to eight holes of 32 × 32 pixels. The final preprocessing step normalizes the RGB channels of the images using the mean and standard deviation derived from the ImageNet dataset. Padding, gap, and hole pixels resulting from transformations are ignored during the experiments. All transformations were implemented using the Albumentations library (https://albumentations.ai/). An overview of this step is shown in Fig. 4.

D. Close-Set Training

Two semantic segmentation networks were used in this work: DeeplabV3 with ResNet-101 backbone [39] and SegFormer [40], pre-trained over the ImageNet [41] and obtained through the Torchvision[1] and Hugging Face[2] library, respectively. Models were trained using a batch size of 32 images for 50 epochs. Early stopping was implemented with a patience of five epochs based on the mIoU values on the validation set. This training approach stops training if the mIoU values do not improve for five consecutive epochs, preventing the model from overfitting on the training data. We used the stochastic gradient descent optimizer with an initial learning rate of 0.02, a weight decay coefficient of 0.0001, and momentum set to 0.09. A cosine annealing schedule adjusted the learning rate, as suggested by [42]. A minimum learning rate of 0.00001 was defined for this process. The trained model with the highest mIoU (see Section IV-G) from the k-fold configuration was selected for the open-set recognition step.

E. Open-Set Training

The open-set classifiers used depend on a closed-set semantic segmentation model. Thus, the model with the highest mIoU computed on the closed-set validation data was chosen for this step. For the POpenDist classifier, no additional training is required as it simply involves applying thresholding to the distance values. In contrast, the GOpenIPCS classifier requires the adjustment of the G parameter to handle the classification of KKCs and UUC pixels, as described in Section III-B2. After this, thresholding is applied to the likelihood outputs. The evaluation of these methods was conducted using the test dataset.

[1]https://pytorch.org/
[2]https://huggingface.co/

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F. Cutoff Thresholds for Open-Set Recognition

All the methods investigated in our study (as compared in Section V) require a cutoff threshold to delineate the limit between KKC and UUC pixels. To undertake a realistic open-set recognition task, these limits are empirically defined based on the KKCs available during the validation data. Notably, no information about the UUCs is used to select the cutoff thresholds for the open-set classifiers. In the case of softmax thresholding (Softmax^T) [2], and OpenMax [9], anomalous probability map (APM) [7], maximum unnormalized logit (MaxLogit) [22], and maximum softmax probability (MSP) [22], the cutoff thresholds are associated with the probability of the network and they are defined as the original papers. The other cutoff thresholds refer to the reconstruction errors of CoReSeg, the prototype distances of POpenDist, and the likelihoods of OpenIPCS and GOpenIPCS, which are within the range [0,∞] and are determined based on true positive rate (TPR) quantiles, according by [2], [13].

G. Evaluation Protocol

This article used the mean intersection-over-union (mIoU) and area under the curve (AUC). The mIoU is a widely used metric to evaluate the performance of closed-set semantic segmentation methods and has recently been applied in the context of OSSS [7], [10], [12]. This metric calculates the ratio between the intersection and the union (IoU) of two sets. For the specific case of image segmentation, the difference between the GT and the prediction pixels is computed. The mIoU measures how accurately the shape of objects is segmented, which is a crucial requirement for image segmentation. The IoU can be computed $\text{IoU} = (\text{TP}_i/\text{TP}_i + \text{FP}_i + \text{FN}_i)$. Then, the mIoU is the average IoU over all classes. Additionally, OSSS results were evaluated using the AUC, a recently used metric to assess OSSS [2], [7], [22]. This metric aids in measuring the model's ability to distinguish between classes, including KKCs and UUCs.

V. Results and Discussion

This section presents the results of our experiments and their discussion. Firstly, Section V-A compares the proposed loss function to the baselines in terms of quantitative metrics, qualitative segmentation, and the embedding space built. Secondly, Section V-B compares the performance of the proposed methods with other state-of-the-art approaches. Subsequently, Section V-C briefly analyzes the proposed online triplets selection. Furthermore, Section V-D presents an evaluation of the proposed method with noise images. Lastly, ablation studies were conducted in Section V-E.

A. Comparison of Loss Functions

This section focuses on the evaluation of the proposed prototypical triplet loss ($\mathcal{L}_{\text{PTL}}$). Additionally, we introduce the combination of $\mathcal{L}_{\text{PTL}}$ and focal loss ($\mathcal{L}_{\text{PTL+FCL}}$) to achieve enhanced results. The performance of these loss functions was evaluated in both, closed-set and OSSS scenarios. The results were compared with three baseline loss functions: cross-entropy ($\mathcal{L}_{\text{CE}}$), focal loss ($\mathcal{L}_{\text{FCL}}$), distance-based cross-entropy combined with variance loss ($\mathcal{L}_{\text{DCE+VL}}$). Table I presents the loss function hyperparameters used for training the model, with values from original papers.

TABLE I
Hyperparameters for the Loss Functions Evaluated in This Article

Ref.	Loss	Hyperparameters
-	$\mathcal{L}_{CE}$	-*
[32]	$\mathcal{L}_{FCL}$	α: 1.0, γ: 2.0
[7]	$\mathcal{L}_{DCE+VL}$	T: 3, λ_{vl}: 0.01
Ours	$\mathcal{L}_{PTL}$	T: 3, λ_g: 1.0, λ_l: 1.0, α_g: 1.5, α_l: 0.5.
Ours	$\mathcal{L}_{PTL+FCL}$	Individual values of the losses were kept.

* Default from https://pytorch.org/.

In Fig. 5, radar charts display the outcomes of our experimental results for closed-set and OSSS, using IoU and mIoU metrics on the Vaihingen and Potsdam Datasets.

Considering the closed-set segmentation, there is a clear similarity of the results, demonstrating that the proposed $\mathcal{L}_{\text{PTL+FCL}}$ performs consistently and competitively with other loss functions found in the literature. Specifically, it achieved the highest average of mIoU of 0.8185 using DeepLabV3 over the Potsdam dataset. Overall, $\mathcal{L}_{\text{PTL+FCL}}$ and $\mathcal{L}_{\text{PTL}}$ are approximately 0.9% and 3.9% lower in average mIoU compared to the best results, respectively. It emphasizes the performance enhancement achieved through the proposed combination with focal loss.

In the context of OSSS, a comparative analysis of loss performance was conducted using two open-set classifiers, GOpenIPCS and POpenDist, proposed in this article. The outcomes are shown in Fig. 5. The cutoff thresholds for the open-set classifiers were defined as described in Section IV-F. The size of the latent space for GOpenIPCS was set to 64, as presented in the ablation study of Section V-E. Regarding the general performance of the results, the proposed functions present a significant advantage over the other baseline functions. Mainly, when combined with POpenDist, a considerable performance enhancement can be observed due to the prototypical anchor distance strategy employed by the methods. Utilizing $\mathcal{L}_{\text{PTL+FCL}}$ for train DeepLabV3, the model achieved an average IoU per UUC of 0.7764 and 0.7862 on the Vaihingen and Potsdam datasets. In the case of training SegFormer, the obtained values were 0.7950 and 0.8024, respectively. Meanwhile, $\mathcal{L}_{\text{PTL}}$ perform slightly worse than $\mathcal{L}_{\text{PTL+FCL}}$, resulting in a reduction of approximately -2.91% in average IoU per UUC. Significant performance enhancements are also evident with GOpenIPCS, highlighting the capacity to build more discriminative embedding vectors through network training using the proposed loss functions. When using $\mathcal{L}_{\text{PTL+FCL}}$ to train DeepLabV3, the model achieved an average IoU per UUC of 0.7250 and 0.7175 on the Vaihingen and Potsdam datasets. In the case of training SegFormer, the obtained values were 0.7001 and 0.7331, respectively. Conversely, $\mathcal{L}_{\text{PTL}}$ perform slightly lower than $\mathcal{L}_{\text{PTL+FCL}}$, resulting in a reduction of approximately -2.72% in average IoU per UUC.

For a comprehensive analysis, Table II provides illustrative visual examples and corresponding label space obtained

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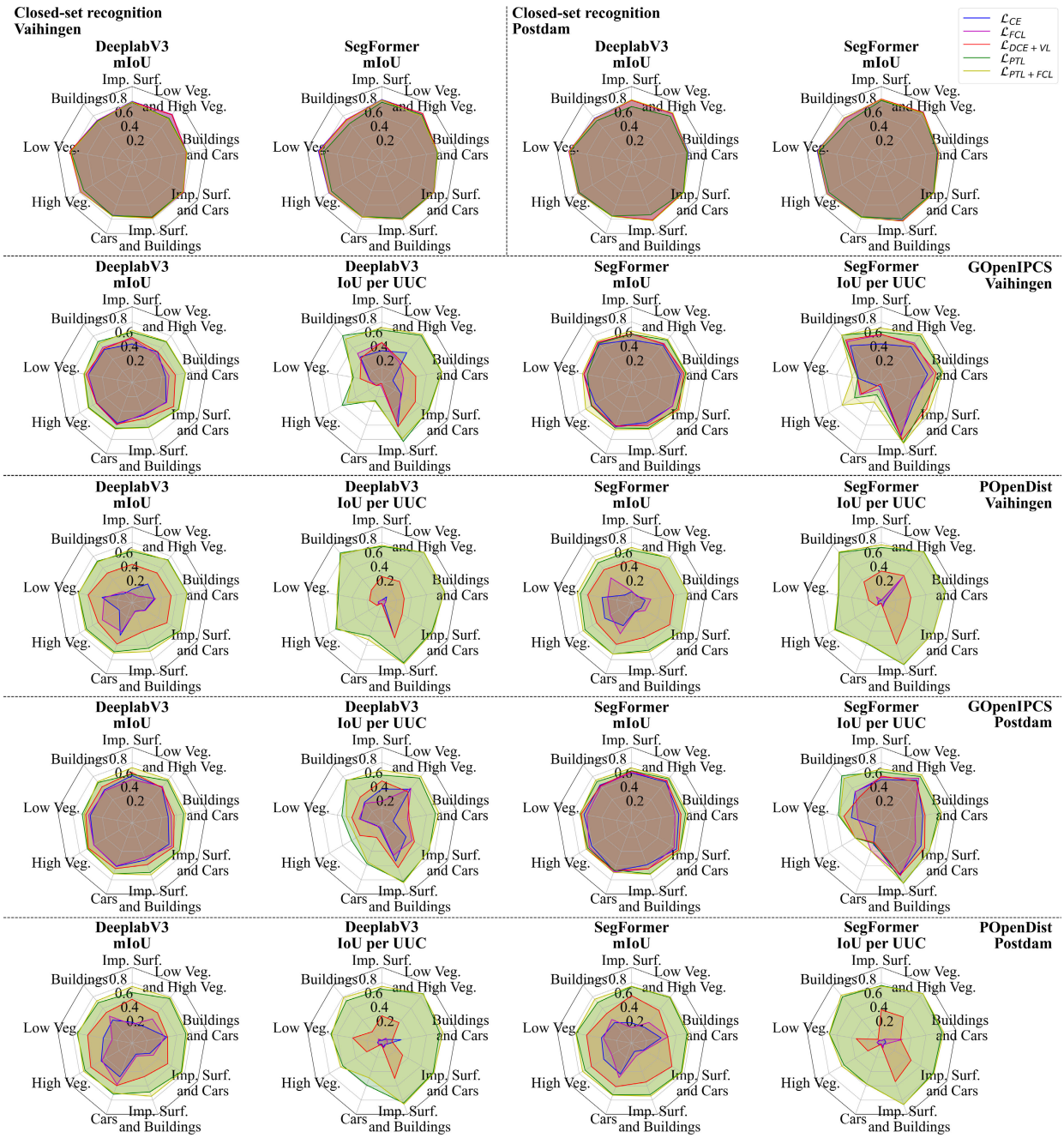


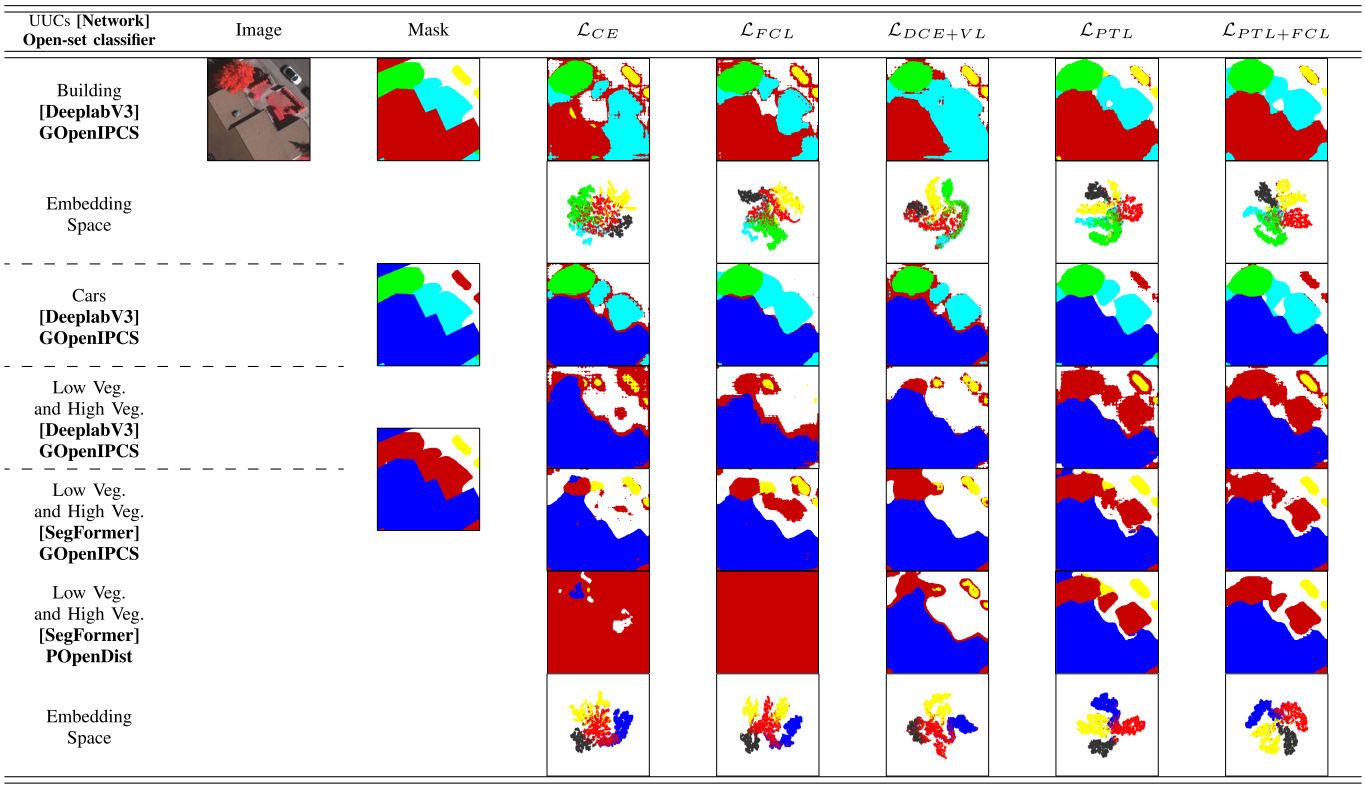
Fig. 5. Radar charts for illustrating the results of closed-set semantic segmentation using the mIoU and IoU metrics. The UUCs are positioned at the vertices of the radar, while the lines represent the compared loss functions in this article. These results were obtained from the DeepLabV3 and SegFormer models on the Vaihingen and Potsdam datasets using GOpenIPCS and POpenDist.

from the open-set experiment on the Vaihingen dataset using GOpenIPCS and POpenDist. A qualitative analysis of the results shows that the proposed loss functions produce outcomes closer to the GT, outperforming other baseline losses. Our approach results in cleaner and more precise segmentations using either GOpenIPCS or POpenDist, alleviating the uncertain boundaries problem that produces UUC artifacts around segmented KKC objects [2], [12]. Furthermore, our approach is more robust to segmenting small objects belonging to the KKC or UUC, alleviating another common problem of semantic segmentation [43], [44]. The main factor for these outcomes can be considerable when investigating the embedding space associated with the proposed loss functions, which exhibits a more controlled embedding space with well-defined decision regions between classes, closely aligning with the "ideal" embedding space, presenting low overlap between the decision boundaries among classes. Contrariwise, baseline losses demonstrate a higher rate of overlap between classes. These results underscore the efficacy of our proposed approach, highlighting that the organized embedding space contributes to the more discriminative capacity of the model and enhances outcomes in OSSS scenarios.

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TABLE II

VISUAL EXAMPLES OF OSSS EXTRACTED FROM THE VAIHINGEN DATASET. THE UUCS ARE THE RED PIXELS. THE EMBEDDING SPACE IS VISUALIZED USING TSNE WITH A PERPLEXITY EQUAL TO 50



B. Comparison With Other Methods

This section compares the proposed methods with the state-of-the-art methods for OSSS on remote sensing images. This comparison encompasses the OpenMax [9], Softmax^T [2], OpenIPCS [2], CoReSeg [13], APM [7], MaxLogit [22], and MSP [22]. All methods considered in this section were trained using the configurations outlined earlier in Section IV, with hyperparameter values consistent with those specified in the original papers. The open-set experiments follow the configurations suggested by Nunes et al. [13].

Table III shows the quantitative results of AUC, mIoU, and IoU per UUCs in the Vaihingen and Potsdam datasets. The results indicate a significant superiority of the proposed approaches over other methods in the literature. Note that the best results are highlighted in bold and were mainly observed using the SegFormer network trained with $\mathcal{L}_{\text{PTL+FCL}}$ and using POpenDist as the open-set classifier. These performance gains are evident in Table IV, where the UUCs are represented by red pixels, demonstrating that the proposed methods provide refined segmentations of the UUCs compared to other baseline methods. Furthermore, despite $\mathcal{L}_{\text{CE}}$ achieving state-of-the-art performance in closed-set semantic segmentation, its efficacy is constrained in open-set scenarios. Its limitations in organizing the embedding space discourage the optimal segmentation of UUCs. Therefore, this article demonstrates the importance of organizing the embedding space to enhance OSSS on remote sensing images.

In addition, our approach was compared with other state-of-the-art methods that use metric learning to improve the OSSS [7], [22]. Although these approaches show performance gains over non-metric learning methods, they still experience performance degradation with similar classes such as "Low Veg." and "High Veg.," as well as with small objects such as those in the "Cars" class. Meanwhile, our approach maintains consistent performance across these challenging scenarios because it explicitly handles the issues of small objects and uncertain boundary regions, unlike other state-of-the-art metric learning approaches for OSSS.

Finally, the results presented in Tables III and IV highlight the potential of our approach to enhance OSSS performance on remote sensing images, mainly including those that are more challenging for OSSS, such as small objects and similar objects [10], [12]. Our proposed $\mathcal{L}_{\text{PTL}}$ effectively organizes the embedding space. It brings samples closer to the prototypical anchor, addresses class imbalance, and considers boundary regions, thus enhancing performance in OSSS. It is worth noting the significant performance improvement achieved when our method is combined with POpenDist. This classifier leverages the prototypical anchor distance strategy without requiring additional training.

C. Analysis of Online Triplets Selection

Triplet selection is a fundamental step for the efficacy of triplet loss learning [30], [33], [34]. Given that the number of triplets grows cubically with the size of the training dataset,

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TABLE III

COMPARISON WITH THE STATE-OF-THE-ART OSSS METHODS. THE NUMBERS AND LETTERS IN [ ] IDENTIFY THE LOSS FUNCTIONS AND NETWORK MODELS USED BY THE OPEN-SET CLASSIFIER: 0—$\mathcal{L}_{CE}$, 1—$\mathcal{L}_{DCE+VL}$, 2—PIXEL CONTRASTIVE LOSS (PC) $\mathcal{L}_{CE+PC}$, 3—$\mathcal{L}_{PTL}$, 4—$\mathcal{L}_{PTL+FCL}$, A—SEGFORMER, B—RN101+DEEPLABV3, C—U-NET, D—DN121+FCN, E—WRN50+FCN, F—DRN50+PSPNET, AND G—RN50+DEEPLABV3

Ref.	Metric Learning?	Open Set Classifier		Imp. Surf. AUC	Imp. Surf. mIoU	Imp. Surf. IoU *UUC*	Buildings AUC	Buildings mIoU	Buildings IoU *UUC*	Low Veg. AUC	Low Veg. mIoU	Low Veg. IoU *UUC*	High Veg. AUC	High Veg. mIoU	High Veg. IoU *UUC*	Cars AUC	Cars mIoU	Cars IoU *UUC*
Vaihingen																		
[2]	×	Softmax^T	[0A]	.835$_{\pm.13}$	.500$_{\pm.23}$	.257	.800$_{\pm.11}$	.515$_{\pm.17}$	.274	.825$_{\pm.16}$	.597$_{\pm.29}$	.114	.806$_{\pm.18}$	.560$_{\pm.33}$	.048	.822$_{\pm.19}$	.607$_{\pm.35}$	.004
[9]	×	OpenMax	[0A]	.841$_{\pm.12}$	.515$_{\pm.22}$	.299	.812$_{\pm.11}$	.524$_{\pm.18}$	.273	.825$_{\pm.16}$	.599$_{\pm.29}$	.128	.812$_{\pm.18}$	.558$_{\pm.34}$	.023	.817$_{\pm.19}$	.604$_{\pm.35}$	.005
[13]	×	CoReSeg	[0C]	.724$_{\pm.13}$	.365$_{\pm.23}$	.081	.697$_{\pm.15}$	.336$_{\pm.25}$	.034	.727$_{\pm.17}$	.420$_{\pm.28}$	.118	.721$_{\pm.16}$	.421$_{\pm.27}$	.083	.787$_{\pm.09}$	.481$_{\pm.27}$	.040
[13]	×	CoReSeg*	[0C]	.712$_{\pm.11}$	.366$_{\pm.17}$	.157	.699$_{\pm.15}$	.361$_{\pm.23}$	.066	.726$_{\pm.17}$	.417$_{\pm.29}$	.071	.716$_{\pm.15}$	.402$_{\pm.26}$	.107	.799$_{\pm.10}$	.516$_{\pm.28}$	.037
[2]	×	OpenIPCS	[0D]	.803$_{\pm.08}$	.481$_{\pm.18}$	.343	.813$_{\pm.05}$	.557$_{\pm.10}$	.447	.798$_{\pm.16}$	.529$_{\pm.27}$	.079	.727$_{\pm.17}$	.447$_{\pm.25}$	.061	.835$_{\pm.14}$	.605$_{\pm.30}$	.100
[2]	×	OpenIPCS	[0E]	.774$_{\pm.13}$	.420$_{\pm.24}$	.153	.781$_{\pm.09}$	.499$_{\pm.15}$	.316	.775$_{\pm.16}$	.500$_{\pm.26}$	.075	.738$_{\pm.16}$	.455$_{\pm.27}$	.070	.831$_{\pm.14}$	.599$_{\pm.29}$	.114
[7]	✓	APM	[1F]	.813$_{\pm.07}$	.563$_{\pm.17}$	.604	.797$_{\pm.06}$	.540$_{\pm.12}$	.560	.829$_{\pm.12}$	.612$_{\pm.24}$	.313	.798$_{\pm.16}$	.538$_{\pm.24}$	.254	.820$_{\pm.16}$	.597$_{\pm.34}$	.024
[22]	✓	MaxLogit	[2G]	.830$_{\pm.12}$	.532$_{\pm.21}$	.339	.785$_{\pm.13}$	.498$_{\pm.25}$	.175	.827$_{\pm.14}$	.596$_{\pm.28}$	.120	.812$_{\pm.16}$	.552$_{\pm.35}$	.124	.819$_{\pm.17}$	.604$_{\pm.35}$	.005
[22]	✓	MSP	[2G]	.824$_{\pm.10}$	.546$_{\pm.18}$	.431	.781$_{\pm.12}$	.508$_{\pm.22}$	.204	.822$_{\pm.13}$	.598$_{\pm.25}$	.179	.806$_{\pm.16}$	.555$_{\pm.34}$	.114	.825$_{\pm.17}$	.606$_{\pm.35}$	.003
Ours	✓	GOpenIPCS	[3B]	.890$_{\pm.05}$	.693$_{\pm.11}$	.663	.896$_{\pm.04}$	.704$_{\pm.09}$	.808	.861$_{\pm.09}$	.643$_{\pm.16}$	.382	.885$_{\pm.05}$	.662$_{\pm.12}$	.607	.867$_{\pm.08}$	.643$_{\pm.24}$	.255
Ours	✓	GOpenIPCS	[4B]	.888$_{\pm.04}$	.690$_{\pm.11}$	.723	.870$_{\pm.04}$	.675$_{\pm.08}$	.747	.852$_{\pm.09}$	.648$_{\pm.17}$	.397	.867$_{\pm.06}$	.670$_{\pm.14}$	.529	.871$_{\pm.08}$	.653$_{\pm.24}$	.267
Ours	✓	GOpenIPCS	[3A]	.881$_{\pm.04}$	.639$_{\pm.16}$	.667	.905$_{\pm.05}$	.675$_{\pm.14}$	.818	.853$_{\pm.11}$	.593$_{\pm.18}$	.318	.861$_{\pm.09}$	.614$_{\pm.17}$	.410	.854$_{\pm.12}$	.639$_{\pm.27}$	.173
Ours	✓	GOpenIPCS	[4A]	.896$_{\pm.04}$	.679$_{\pm.14}$	.716	.894$_{\pm.03}$	.714$_{\pm.09}$	.817	.860$_{\pm.09}$	.659$_{\pm.17}$	.389	.880$_{\pm.05}$	.700$_{\pm.12}$	.598	.874$_{\pm.08}$	.663$_{\pm.24}$	.275
Ours	✓	POpenDist	[3B]	.907$_{\pm.06}$	.677$_{\pm.14}$	.742	.904$_{\pm.04}$	.716$_{\pm.11}$	.857	**.900**$_{\pm.03}$	.719$_{\pm.11}$	.583	.906$_{\pm.04}$	.696$_{\pm.12}$	.701	.881$_{\pm.06}$	.690$_{\pm.16}$	.468
Ours	✓	POpenDist	[4B]	.902$_{\pm.04}$	.704$_{\pm.12}$	.756	.886$_{\pm.04}$	.704$_{\pm.11}$	.841	.888$_{\pm.04}$	.721$_{\pm.11}$	**.601**	.888$_{\pm.04}$	.715$_{\pm.11}$	.682	.884$_{\pm.06}$	.707$_{\pm.14}$	.519
Ours	✓	POpenDist	[3A]	**.909**$_{\pm.05}$	.692$_{\pm.14}$	.728	**.912**$_{\pm.04}$	.687$_{\pm.16}$	.865	.900$_{\pm.04}$	.688$_{\pm.13}$	.574	**.912**$_{\pm.04}$	.712$_{\pm.04}$	.702	**.897**$_{\pm.04}$	**.723**$_{\pm.12}$	**.569**
Ours	✓	POpenDist	[4A]	.907$_{\pm.04}$	**.729**$_{\pm.11}$	**.763**	.903$_{\pm.03}$	**.735**$_{\pm.10}$	**.873**	.894$_{\pm.05}$	**.727**$_{\pm.11}$	.580	.899$_{\pm.04}$	**.740**$_{\pm.11}$	**.718**	**.897**$_{\pm.04}$	.723$_{\pm.13}$	.562
Postdam																		
[2]	×	Softmax^T	[0A]	.835$_{\pm.14}$	.552$_{\pm.20}$	.199	.813$_{\pm.18}$	.537$_{\pm.30}$	.006	.849$_{\pm.12}$	.620$_{\pm.20}$	.283	.835$_{\pm.02}$	.599$_{\pm.35}$	.022	.835$_{\pm.19}$	.639$_{\pm.36}$	.010
[9]	×	OpenMax	[0A]	.834$_{\pm.13}$	.573$_{\pm.18}$	.258	.812$_{\pm.17}$	.541$_{\pm.29}$	.023	.854$_{\pm.11}$	.634$_{\pm.19}$	.319	.826$_{\pm.18}$	.605$_{\pm.32}$	.078	.838$_{\pm.19}$	.641$_{\pm.36}$	.010
[13]	×	CoReSeg	[0C]	.699$_{\pm.12}$	.344$_{\pm.20}$	.041	.715$_{\pm.12}$	.388$_{\pm.21}$	.230	.715$_{\pm.14}$	.368$_{\pm.24}$	.024	.713$_{\pm.15}$	.409$_{\pm.27}$	.014	.821$_{\pm.04}$	.517$_{\pm.25}$	.078
[13]	×	CoReSeg*	[0C]	.728$_{\pm.14}$	.404$_{\pm.25}$	.050	.731$_{\pm.11}$	.422$_{\pm.21}$	.204	.731$_{\pm.15}$	.397$_{\pm.25}$	.034	.725$_{\pm.15}$	.434$_{\pm.28}$	.016	.827$_{\pm.05}$	.526$_{\pm.26}$	.076
[2]	×	OpenIPCS	[0D]	.820$_{\pm.09}$	.536$_{\pm.14}$	.307	.835$_{\pm.05}$	.588$_{\pm.09}$	.443	.796$_{\pm.14}$	.502$_{\pm.26}$	.087	.834$_{\pm.15}$	.589$_{\pm.29}$	.094	.872$_{\pm.07}$	.652$_{\pm.22}$	.275
[2]	×	OpenIPCS	[0E]	.777$_{\pm.12}$	.461$_{\pm.20}$	.151	.796$_{\pm.09}$	.501$_{\pm.17}$	.236	.744$_{\pm.14}$	.434$_{\pm.22}$	.077	.799$_{\pm.14}$	.553$_{\pm.27}$	.075	.846$_{\pm.10}$	.621$_{\pm.25}$	.183
[7]	✓	APM	[1F]	.831$_{\pm.10}$	.592$_{\pm.16}$	.564	.793$_{\pm.11}$	.515$_{\pm.18}$	.192	.827$_{\pm.12}$	.576$_{\pm.23}$	.432	.825$_{\pm.13}$	.606$_{\pm.25}$	.257	.823$_{\pm.15}$	.598$_{\pm.32}$	.048
[22]	✓	MaxLogit	[2G]	.821$_{\pm.11}$	.553$_{\pm.17}$	.248	.791$_{\pm.14}$	.502$_{\pm.25}$	.063	.825$_{\pm.11}$	.569$_{\pm.21}$	.328	.821$_{\pm.11}$	.588$_{\pm.31}$	.106	.821$_{\pm.15}$	.590$_{\pm.33}$	.033
[22]	✓	MSP	[2G]	.812$_{\pm.08}$	.563$_{\pm.14}$	.381	.779$_{\pm.13}$	.504$_{\pm.22}$	.179	.820$_{\pm.09}$	.580$_{\pm.16}$	.432	.818$_{\pm.15}$	.595$_{\pm.29}$	.118	.823$_{\pm.16}$	.591$_{\pm.33}$	.034
Ours	✓	GOpenIPCS	[3B]	.883$_{\pm.05}$	.641$_{\pm.07}$	.631	.902$_{\pm.04}$	.696$_{\pm.03}$	.737	.874$_{\pm.06}$	.669$_{\pm.11}$	.536	.881$_{\pm.07}$	.683$_{\pm.13}$	.463	.893$_{\pm.04}$	.716$_{\pm.10}$	.574
Ours	✓	GOpenIPCS	[4B]	.897$_{\pm.03}$	.728$_{\pm.06}$	.697	.889$_{\pm.01}$	.710$_{\pm.02}$	.728	.857$_{\pm.07}$	.645$_{\pm.13}$	.475	.864$_{\pm.08}$	.673$_{\pm.17}$	.388	.895$_{\pm.04}$	.716$_{\pm.11}$	.550
Ours	✓	GOpenIPCS	[3A]	.905$_{\pm.04}$	.689$_{\pm.08}$	.676	.917$_{\pm.04}$	.712$_{\pm.08}$	.813	.893$_{\pm.06}$	.693$_{\pm.10}$	.575	.878$_{\pm.10}$	.668$_{\pm.16}$	.399	.876$_{\pm.08}$	.665$_{\pm.22}$	.289
Ours	✓	GOpenIPCS	[4A]	.908$_{\pm.03}$	.724$_{\pm.06}$	.712	.902$_{\pm.02}$	.737$_{\pm.04}$	.768	.879$_{\pm.06}$	.688$_{\pm.12}$	.555	.874$_{\pm.09}$	.690$_{\pm.17}$	.397	.889$_{\pm.06}$	.695$_{\pm.18}$	.396
Ours	✓	POpenDist	[3B]	.898$_{\pm.05}$	.667$_{\pm.09}$	.711	.906$_{\pm.04}$	.696$_{\pm.04}$	.729	.911$_{\pm.04}$	.737$_{\pm.06}$	.684	.903$_{\pm.05}$	.719$_{\pm.09}$	.597	.903$_{\pm.03}$	.719$_{\pm.10}$	**.589**
Ours	✓	POpenDist	[4B]	.895$_{\pm.05}$	.748$_{\pm.06}$	.749	.903$_{\pm.04}$	.727$_{\pm.04}$	.786	.895$_{\pm.03}$	.727$_{\pm.07}$	.672	.901$_{\pm.03}$	.740$_{\pm.08}$	**.628**	.894$_{\pm.04}$	.702$_{\pm.14}$	.489
Ours	✓	POpenDist	[3A]	**.915**$_{\pm.03}$	.746$_{\pm.06}$	**.761**	**.916**$_{\pm.04}$	.699$_{\pm.10}$	.798	**.918**$_{\pm.04}$	.744$_{\pm.07}$	.696	.904$_{\pm.06}$	.711$_{\pm.10}$	.572	.898$_{\pm.03}$	.725$_{\pm.11}$	.572
Ours	✓	POpenDist	[4A]	.909$_{\pm.02}$	**.751**$_{\pm.06}$	.753	.911$_{\pm.02}$	**.756**$_{\pm.05}$	**.826**	.908$_{\pm.03}$	**.748**$_{\pm.07}$	**.700**	**.904**$_{\pm.05}$	**.747**$_{\pm.09}$	.605	**.909**$_{\pm.11}$	**.732**$_{\pm.02}$	.574

* CoReSeg has been trained with 100 epochs with a patience of 10.

triplet selection becomes indispensable for achieving efficient training with triplet loss. However, in practical applications, the training process often proves highly sensitive to the triplet selection, demonstrating minimal convergence when utilizing randomly chosen triplets. Conversely, selecting the most challenging triplets can lead to undesirable outcomes, including convergence difficulties and suboptimal local minima [33]. To tackle these challenges, we introduced an online mining selection of triplets based on matrix confusion rules, which are further explained in Section III-A. This section will provide a complementary analysis of the proposed approach, clarifying its significance in the semantic segmentation context.

The global attention mechanics aims to bring pixels closer to their corresponding prototypical anchor. To achieve this, this work randomly selects the TP as triple positives. Also, we want pixels from other classes to be distanced. In this sense, we focus on FP pixels, the hardest pixels to distinguish, as they are generally located close to the boundaries of objects and are continually classified as belonging to the class of interest. When choosing these triplets, it is ensured that the correctly classified pixels are close to the prototypical anchor while distancing incorrectly classified pixels.

The local attention mechanics was built primarily to deal with pixels erroneously classified as negative (FN) and generally located on the boundaries between objects, as seen in Table V. Thus, we define the TP pixels as anchors and the FN pixels as positive triplets, with the objective to make the embeddings of these pixels more similar. For the negative triplets, the TN and FP are randomly selected. In this case, we do not mainly focus on distancing the FP pixels, as this makes the training process very difficult and discourages the model from converging.

The combination of these two strategies for triplet selection, along with the average of triplet losses per class used by $\mathcal{L}_{PTL}$, has produced significant improvements in both quantitative metrics and qualitative assessments. These outcomes can be seen in previous Sections V-B and V-B. Examining Table V, it becomes apparent that the uncertainty boundary

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TABLE IV

VISUAL EXAMPLES OF OSSS EXTRACTED FROM THE VAIHINGEN DATASET. THE UUCS ARE REPRESENTED BY THE RED PIXELS. THE LOSS FUNCTIONS AND NETWORK MODELS CORRESPOND TO THOSE PRESENTED IN TABLE III. THE PROPOSED GOPENIPCS AND POPENDIST SELECTED USE SEGFORMER AND $\mathcal{L}_{\text{PTL+FCL}}$, CORESEG REPRESENTS THE MODEL TRAINED FOR 100 EPOCHS, AND OPENIPCS IS THE MODEL USES DN121+FCN

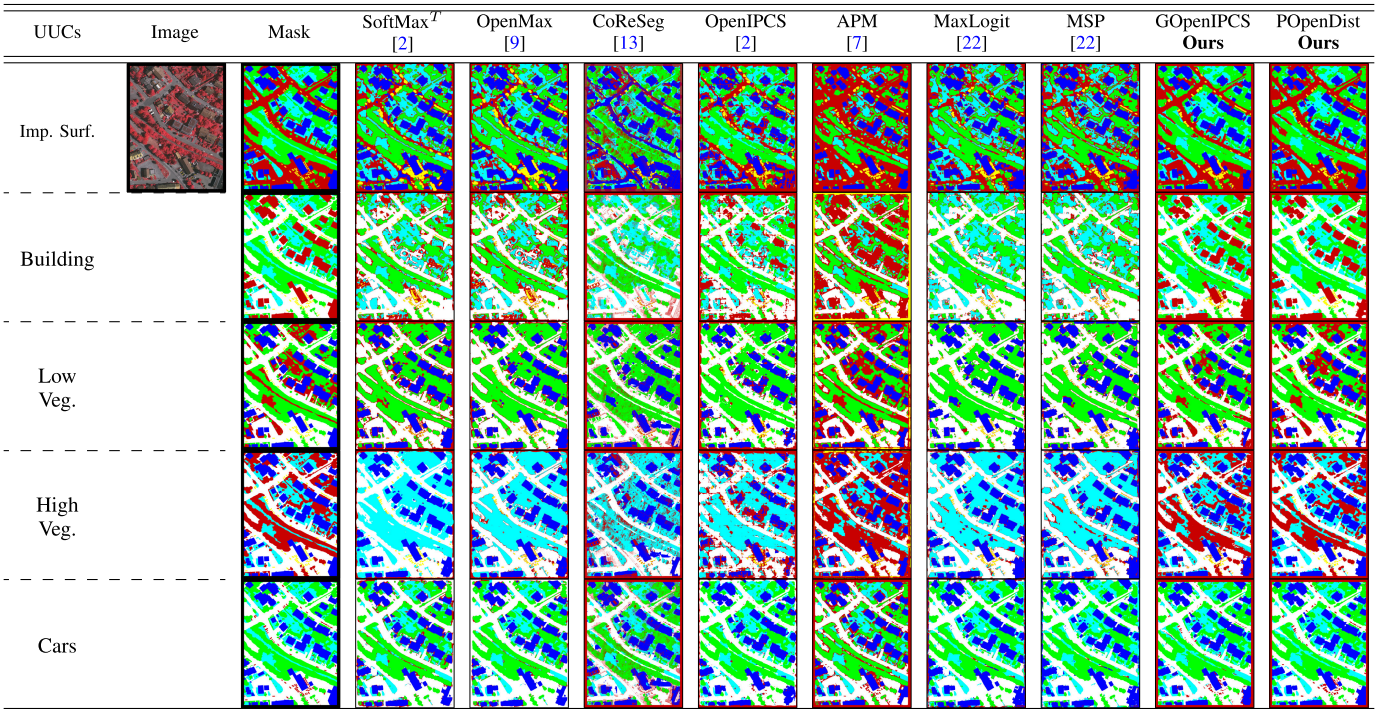
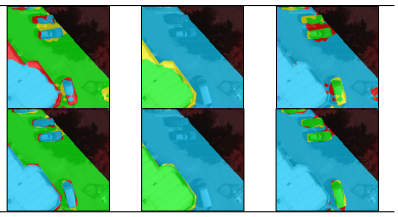


TABLE V

PIXEL-LEVEL CONFUSION MATRIX PER CLASS IN TRAINING BATCH. THE OUTPUTS WERE OBTAINED USING $\mathcal{L}_{\text{PTL}}$ AND DEEPLABV3 ON THE VAIHINGEN DATASET WITH THE FOLLOWING UUC: LOW AND HIGH VEG. THE PIXEL-LEVEL CONFUSION MATRIX RULES ARE PRESENTED: TP—GREEN, FP—RED, TN—BLUE, AND FN—YELLOW. THE OTHER PIXELS REFER TO UUC AND ARE IGNORED BY $\mathcal{L}_{\text{PTL}}$ DURING TRAINING

Epoch	Imp. Surf.	Buildings	Cars
0			
4			

regions diminish throughout the training process. Furthermore, in Table II, consistent and improved results are evident in the segmentation of borders and small objects when employing both $\mathcal{L}_{\text{PTL}}$ and $\mathcal{L}_{\text{PTL+FCL}}$.

D. Robustness of Random Noise

In order to evaluate the robustness of our approaches to random noise, we apply Gaussian noise with zero mean and variance $\sigma \in [0.01, 0.2]$ to 50% of the test data from the Vaihingen dataset. Fig. 6 presents mIoU under different noise variations for the various methods compared in this work. Overall, one can observe that all methods are susceptible to

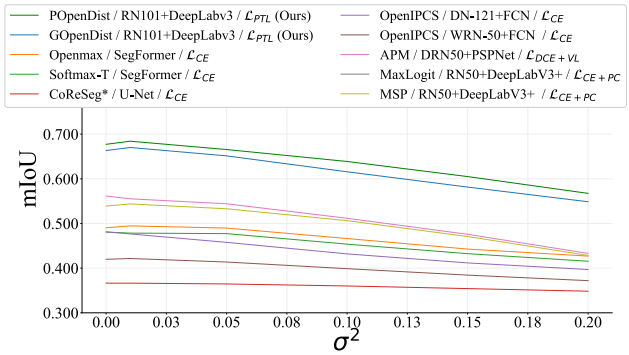


Fig. 6. OSSS results with Gaussian noise applied to the Vaihingen test dataset with UUC: low and high veg.

noise, leading to a performance decline proportionate to the increase in noise variation in the images. Despite the impact of increased noise variation, the proposed GOpenIPCS and POpenDist still demonstrate superior performance compared to the approaches examined in this work. Introducing more random noise during training data may mitigate this degradation problem, but open-set recognition approaches relying on cutoff thresholds can be more affected by noisy pixels in KKCs.

E. Ablation Studies

We conducted an ablation study to determine the optimal global (α_g) and local margin (α_g) values to be applied to the proposed $\mathcal{L}_{\text{PTL}}$ in our experiments. We performed an exhaustive search on a set of margin values (as shown in Table VI)

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TABLE VI

ABLATION STUDY OF THE COMBINATIONS VALUES OF THE GLOBAL AND LOCAL MARGINS OF PROPOSED $\mathcal{L}_{\text{PTL}}$. CLOSED-SET SEMANTIC SEGMENTATION RESULTS UTILIZED THE DEEPLABV3 ON THE VAIHINGEN DATASET WITH UUC: LOW AND HIGH VEG. IGNORED IN THIS SETTING. RESULTS ARE PRESENTED IN TERMS OF MIOU

global (α_g)	local (α_l) 0.1	0.2	**0.5**	1.0
0.5	.709	.785	.747	.731
1.0	.555	.590	.731	.734
1.5	.704	.753	**.801**	.719
2.0	.749	.767	.765	.735

TABLE VII

ABLATION STUDY TO INVESTIGATE THE EFFECTS OF THE SIZE OF LATENT SPACE USED IN GOPENIPCS. OSSS RESULTS USING DEEPLABV3 AND SEGFORMER ON THE VAIHINGEN DATASET WITH UUC: LOW AND HIGH VEG. RESULTS ARE PRESENTED IN TERMS OF MIOU

Network	Size of latent space (L) 8	16	32	64
DeepLabv3	.707	.701	.701	**.710**
SegFormer	.718	.729	.731	**.739**

TABLE VIII

ABLATION STUDY OF THE PROPOSED $\mathcal{L}_{\text{PTL}}$ AND $\mathcal{L}_{\text{PTL+FCL}}$. THE CHECK MARK (✓) INDICATES THE COMPONENTS USED FOR TRAINING THE NETWORK. OSSS RESULTS FOR THE VAIHINGEN DATASET USING DEEPLABV3 AND GOPENIPCS WITH UCC: LOW AND HIGH VEG. RESULTS ARE PRESENTED IN TERMS OF MIOU

UUC [Network]	$\mathcal{L}_{PTL}^{Global}$	$\mathcal{L}_{PTL}^{Local}$	$\mathcal{L}_{FCL}$	mIoU GOpenIPCS	POpenDist
Low and High Veg. [DeepLabV3]	✓			.617	.685
		✓		.699	.493
	✓	✓		.701	**.738**
			✓	.489	.114
	✓	✓	✓	**.707**	.731

and selected the combination of margins that resulted in the best performance. The global margin was defined as 1.5, and the local margin as 0.5. In all experiments presented in this work, we used the proposed $\mathcal{L}_{\text{PTL}}$ with these margin settings.

For GOpenDist, we conducted an ablation study to determine the size of the latent space vector, as illustrated in Fig. 3. Our experiments considered [8, 16, 32, 64] vector sizes. The results presented in Table VII indicate that performance improves with an increase in the size of the latent space vectors. However, the performance gain is also relatively small, so we limit the experiments to this set of values and set the latent space value equal to 64 in our experiments.

At last, ablation experiments were conducted carefully to examine in depth the effect of $\mathcal{L}_{\text{PTL}}^{\text{Global}}$ and $\mathcal{L}_{\text{PTL}}^{\text{Local}}$ on the $\mathcal{L}_{\text{PTL}}$, as illustrated in Table VIII. The outcomes indicate that both loss functions have distinct discriminative capabilities. However, when combined, they provide superior performance compared with their individual implementations. This underscores the effectiveness of integrating these function losses in the proposed $\mathcal{L}_{\text{PTL}}$ and $\mathcal{L}_{\text{PTL+FCL}}$.

VI. CONCLUSION

Semantic segmentation under open-set conditions still represents a significant and complex challenge in computer vision. It is crucial to ensure the reliability and robustness of such systems in handling and interpreting UUCs, especially in remote sensing image mapping and monitoring applications. Hence, it is essential to develop effective open-set recognition methods to address the challenges of UUCs.

This article introduced a novel loss function called the prototypical triplet loss, which uses a prototype representation and metric learning to enhance the recognition of open-set classifiers for semantic segmentation. The proposed loss function improves class discriminability, increasing the open space between classes in the embedding space. Jointly proposes a simple and efficient online triple selection to ensure convergence of semantic segmentation networks. Also, our approach handles problems related to class imbalance and uncertainties present in boundary regions between objects. Additionally, two open-set classifiers based on principal components and prototypical distance were proposed, which led to significant improvements when combined with the proposed loss function.

Our experimental protocols showed that the proposed prototypical triplet loss is competitive with other loss functions in closed-set scenarios. In open-set scenarios, our proposed loss function shows significant performance improvements compared to loss function baselines. Moreover, our approach uses the proposed open-set methods that perform significantly better than the state-of-the-art open-set classifiers used in OSSS for remote sensing images and metric learning approaches proposed for OSSS, demonstrating the effectiveness of the proposed method.

Future works will focus on increasing intra-class compactness and inter-class separation since this issue is crucial to enhancing open-set recognition. Finally, we expect to foster more researchers to explore the potential of the proposed approach to boost OSSS on remote sensing images and other similar real-world applications.

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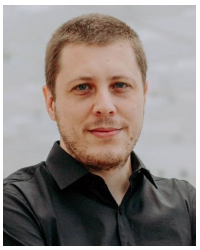
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